Faculty of Science, Engineering and Built Environment

SEJ441 Engineering Project A

Deakin University Unit Guide

Trimester 2, 2025

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Welcome

In this unit, students will develop a detailed plan and academic justification for an engineering research project, supported by significant preliminary or proof-of concept work. Students will learn and use research methodologies and project management approaches to prepare them for tackling an extended engineering project from conception through to realisation. While students are expected to take ownership of their project, they should work closely with their academic supervisor to clearly define the scope, objectives, justification and methodology for the project. Students should also consider requirements of other key stakeholders, as well as safety, risk and ethical factors in the design of their project. Students will develop advanced theoretical and technical knowledge in their project area during the development and refinement of their methodology. Students will also design and conduct initial project work to demonstrate that the methodology or initial concepts are viable and that project objectives can be met.
The project designed in this unit and the preliminary project work undertaken in this unit will be continued and completed in SEJ446.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Ellen Moon
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Email: ellen.moon@deakin.edu.au
Phone: +61 3 522 78656
Other members of the team and how to contact them

Either before the trimester begins, or during weeks 0 and 1 of trimester, you will need to find a project and supervisor.

Information on how to do this is found online in both the Engineering Course Hub and the SEJ441 unit site. Your supervisor will assist you during the unit with understanding, designing and conducting the project, as well as preparing assessment. You may also be working with Technical Officers in the area of your project.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).

In previous versions of this unit, students have told us that these aspects of the unit have helped them to achieve the learning outcomes:

- regularly meeting with their supervisor
- carefully going through the learning resources to understand requirements and expectations
- attending or watching the weekly Zoom seminars.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- provided more recent examples of previous students' work that is discussed in Zoom seminars.
- the project management assessment has been embedded across the trimester rather than wholly left to the end of trimester to promote the regular use of project management tools and techniques, and better align with industry practice.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Demonstrate a competent understanding of relevant background literature and potential approaches to undertake the project in justifying the need for the project including its engineering significance, and developing a relevant and achievable aim and project approach.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO5: Problem solving
ULO2	Demonstrate effective management of an engineering project, including (but not limited to) considerations such as project scheduling, risk management, stakeholder engagement, professional conduct, safety, sustainability, and ethical contexts.	GLO6: Self-management GLO8: Global citizenship
ULO3	Effectively and convincingly communicate and defend deep understanding of technical content, project objectives and project-design reasoning in an oral presentation.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication
ULO4	Produce, analyse, interpret and discuss preliminary results to validate and/or justify the proposed project objectives and methodology	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO4: Critical thinking
ULO5	Act on stakeholder feedback and learnings from preliminary results to develop a detailed engineering design or research justification that identifies and critically appraises recent developments and emerging issues in the research area, and applies engineering approaches informed by research literature to develop a detailed methodology.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO4: Critical thinking GLO5: Problem solving

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit students must achieve a mark of at least 50% in both the project management

portfolio (AT3) and the academic justification (AT4).

Brief summary of the hurdle requirements	Rationale
To be eligible to obtain a pass in this unit students must achieve a mark of at least 50% in both the project management portfolio (Assessment 3) and the academic justification (Assessment 4).	The project undertaken in SEJ441 is continued in SEJ446, and thus students must demonstrate they have developed their project to a satisfactory level in the academic justification and project management portfolio to meet the ULOs and be able to continue into SEJ446.

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Initial proposal**
Brief description of assessment task	This is an individual assessment task.
	The initial proposal is an individual report that defines the initial aims, scope, stakeholder and resource requirements for the project. Students will need to justify the significance and benefit of the project and demonstrate a clear understanding of key considerations relevant to the project design and management.
Detail of student output	Students will be required to submit a written proposal report.
Grading and weighting (% total mark for unit)	25% - marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	This task directly assesses ULO1 and ULO2. By preparing a project proposal, supported by literature, to tackle a problem with engineering significance, students demonstrate ULO1. By including considerations such as sustainability, ethics, safety in the project design, students demonstrate ULO2.
This task assesses your achievement of these Graduate Learning Outcome(s)	Students will need to demonstrate and communicate an understanding of the key technical information in their project area (GLO1 and GLO2), including the sourcing and interpretation of relevant research literature (GLO3), and developing appropriate aims, objectives and approach (GLO5). Students also need to demonstrate effective project planning and management considerations (GLO6 and GLO8).
How and when you will receive feedback on your work	Each student is supported by one academic supervisor and progressive feedback will be given by the academic supervisor.
When and how to submit your work	Monday 4 August 2025 (Week 5) by 8pm (AEST) All written reports/proposals should be submitted electronically through the relevant dropbox in the unit site.

Summative assessment task 2

Title	**Oral presentation**

Brief description of assessment task	This is an individual assessment task. Students will deliver an oral presentation supported by slides or other visuals where the project aims, justification and initial results should be clearly and professionally communicated to a technical audience. Students must demonstrate that they possess a thorough understanding of the rationale behind their project design, and should present initial preliminary results or outcomes from a feasibility study.
Detail of student output	10 minute presentation followed by 10 minute question/answer (via live videolink).
Grading and weighting (% total mark for unit)	20% - graded
This task assesses your achievement of these Unit Learning Outcome(s)	This task directly assesses ULO1, ULO3 and ULO4. By communicating a project proposal, supported by literature, to tackle a problem with engineering significance, students demonstrate ULO1. By including and discussing preliminary results, students demonstrate ULO4. By delivering the proposal in the format of an oral presentation, and by justifying and defending their project design in response to assessors questions, students demonstrate ULO3.
This task assesses your achievement of these Graduate Learning Outcome(s)	Students will be assessed on their depth of technical knowledge (GLO1) and the manner in which they use it to defend their project approach (GLO5), as well as their oral presentation skills (GLO2).
How and when you will receive feedback on your work	Students should seek feedback from their academic supervisor before the final presentation.
When and how to submit your work	Friday 12 September 2025 (Week 9). Specific times will be advised through the unit site.

Summative assessment task 3

Title	**Project management portfolio**
Brief description of assessment task	This is an individual assessment task. Over the course of the trimester, students will compile an online portfolio of evidence of appropriate project planning and management activities, such as logbooks, meeting minutes, project timelines, project safety assessments, ethical considerations, and project budget and resources. As part of the portfolio students are required to provide written reflections on their progress and project management. Templates and suggested due dates are provided in the online portfolio to support record keeping and appropriate progress. The portfolio culminates in a final project plan that must contain detailed project-relevant tasks, resources and timings that incorporate SEJ446.
Detail of student output	An online portfolio containing regular evidence of a range of project management considerations and a final project plan.
Grading and weighting (% total mark for unit)	20% - marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	This task directly assesses ULO2. By preparing a portfolio of evidence of project progress, project planning, stakeholder engagement, and reflections, students demonstrate ULO2.
This task assesses your achievement of these Graduate Learning Outcome(s)	The project management portfolio will assess students project progress and self management skills (GLO6). Students will also be assessed on how well they identify and address relevant safety, ethical, risk and other professional considerations (GLO8).

How and when you will receive feedback on your work	Students are expected to meet supervisors on a regular basis throughout the trimester and keep accurate minutes or logs of meetings, supervisor feedback and follow-up actions. Students will receive feedback on their portfolios during the trimester from the unit team.
When and how to submit your work	Final submission - Monday 29 September 2025 (Week 12 - study week) by 8pm (AEST). The portfolio should be submitted electronically through the relevant tool in the unit site.

Summative assessment task 4

Title	**Academic justification**
Brief description of assessment task	This is an individual assessment task. The academic justification is a detailed report that contains fully justified project aim, objectives and scope, literature review and methodology. Students must also present and discuss their preliminary results and/or outcomes of their proof-of-concept or feasibility study, and how this has influenced the project design. The academic justification must also directly address how any feedback resulting from the initial proposal or presentation has been incorporated into the project design.
Detail of student output	Students will be required to submit a technical report that addresses provided criteria.
Grading and weighting (% total mark for unit)	35% - graded
This task assesses your achievement of these Unit Learning Outcome(s)	This task directly assesses ULO4 and ULO5. By developing a rigorous and detailed project proposal that incorporates feedback from previous assessments, students demonstrate ULO5. By including a thorough analysis of preliminary results to justify their project approach, students demonstrate ULO4.
This task assesses your achievement of these Graduate Learning Outcome(s)	The academic justification will assess student's ability to clearly communicate (GLO2) the depth and breadth of their technical knowledge in the project area (GLO1), their understanding and critical analysis of relevant research literature (GLO3 and GLO4), their use of literature and technical knowledge to propose justified objectives and methodology (GLO5), as well as a critical analysis of their preliminary results and assessor feedback (GLO4) and how these have influenced project design (GLO5)
How and when you will receive feedback on your work	Students are required to incorporate feedback from assessment tasks 1 and 2 into this report. Progressive feedback should also be sought by the student from the academic supervisor.
When and how to submit your work	Monday 6 October 2025 (End-of-unit assessment period) by 8pm (AEST). All written reports should be submitted electronically through the relevant dropbox in the unit site.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 1 hour online seminar per week.
In addition to Project work undertaken on campus, students are required to maintain a minimum of one hour fortnightly contact with their academic supervisor face-to-face. Refer to the unit site for further details.

Educator-facilitated (scheduled) learning activities - online unit enrolment

1 x 1 hour online seminar per week.
Students are required to maintain a minimum of one hour fortnightly contact with their academic supervisor face-to-face or online. Refer to the unit site for further details.

Typical study commitment

Students will on average spend 300 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.
This will include educator guided online learning activities within the unit site.

Note:

- the oral presentation for both campus and online students will be via video conference (not physical attendance on campus); and
- there is a 1-hour weekly online Zoom seminar for online and campus students.

For campus and online students learning and assessment activities include meetings with supervisor/s and technical staff, literature research, designated project activities and assessment tasks.

Each student will have an individual project and be supervised by an academic supervisor.

Students are the project leaders, and as such, it is their responsibility to ensure that all deadlines are met, all training, equipment, materials are booked and ordered in time, and that meetings with required personnel (such as supervisors or technical officers) are arranged well in advance.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.
Unit learning resources

Your unit learning resources can be accessed from your unit site.
There is no prescribed text. Unit materials are provided via the unit site. This includes unit topic readings and references to further information.
Essential learning resources

Students will be provided with support material in the form of video content, library links, readings and examples to assist them in understanding expectations, undertaking their projects and producing assessment tasks. Students will also be

required to locate and use other literature specific to their project topic through library databases and web searches.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

Useful learning sources will be discussed, as appropriate, with your supervisor and/or posted on the unit site during the trimester.

Key dates for this study period

Trimester 2 teaching period begins	Monday 7 July 2025
Census Date	Friday 15 August 2025
Intra-trimester break	Monday 11 August - Sunday 17 August 2025
Trimester 2 teaching period ends	Friday 26 September 2025
Study period (end-of-unit assessment/examination preparation period)	Monday 29 September - Friday 3 October 2025
End-of-unit assessment/examination period	Monday 6 October - Friday 17 October 2025
Inter-trimester break (the period between trimesters)	Monday 20 October - Friday 31 October 2025
Unit results released	Friday 31 October 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Assessment due date
1	7 July 2025	Discussions with supervisor to understand the context of the project. Set up of meeting schedule with supervisor for the trimester.	
2	14 July 2025	General background research & reading on project topic. Complete week 2 project management portfolio entry.	
3	21 July 2025	Specific research & reading (prioritising peer reviewed literature) related to project topic. Formulate project aim & identify potential project approaches.	
4	28 July 2025	Refine project approach through consultation with stakeholders (incl. technical officers). Prepare initial proposal. Complete week 4 project management portfolio entry.	
5^	4 August 2025	Start arranging training for labs and equipment needed for your project, order materials, etc. Continue reading journals and peer reviewed or more specific literature related to the project.	**Assessment 1: Initial proposal** due 4 August
6	18 August 2025	Complete lab/equipment training & inductions and draft risk assessment. Complete week 6 project management portfolio entry.	
7	25 August 2025	Commence proof of concept tasks. Increase literature reading to support detailed literature review.	

Week	Commencing	Topic	Assessment due date
8	1 September 2025	Continue proof of concept tasks. Prepare presentation. Complete week 8 project management portfolio entry.	
9	8 September 2025	Finalise project safety paperwork. Continue proof of concept tasks. Deliver presentation.	**Assessment 2: Oral presentation** due 12 September
10	15 September 2025	Complete proof of concept tasks. Complete week 10 project management portfolio entry. Use feedback from initial proposal and presentation to further develop literature understanding, aims, objectives, and methodology. Begin final project plan tasks within project management portfolio.	
11#	22 September 2025	Continue to implement assessor feedback to refine academic justification. Complete final project plan tasks within project management portfolio.	
12	29 September 2025	Finalise and submit project management portfolio.	Study period **Assessment 3: Project management portfolio** due 29 September
13-14	6 October 2025	Finalise and submit academic justification.	End-of-unit assessment and exam period **Assessment 4: Academic justification** due 6 October 2025

^ Intra-trimester break: **Monday 11 August – Sunday 17 August 2025**

AFL Grand Final Eve public holiday: **Friday 26 September 2025** - University closed